

BTCS502

III Semester Examination, December 2019

B.Tech. /B.Tech. +MBA/B.Tech. +M.Tech. [IT]**Operating System**

Choice Based Credit System (CBCS)

Time: 3 Hrs.**Maximum Marks : 60****Minimum Pass Marks: 24**

- Note:* (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 (2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 (3) Draw the neat diagram, wherever necessary.
 (4) Assume suitable data, wherever necessary.

- Q.1(A)** What is meant by Operating System? Explain its three main purposes. **03**
(B) What are the two differences between user-level threads and kernel level threads? **03**
(C) What is the purpose of the system calls? Also described the major groups of system calls in detail. **06**

OR

Discuss Multi threading Model. Explain the resources used when thread is created.

- Q.2(A)** What is meant by schedulers and context switching? **03**
(B) Explain dispatcher and dispatch latency. **03**
(C) Consider the following set of processes, with the length of the CPU-burst time given in milliseconds: **06**

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The Processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

- a) Draw Gantt charts for illustrating the execution of these processes using FCFS, SJF, nonpreemptive priority (a smaller priority number implies a higher priority), and RR (quantum=1) scheduling.

Contd.....

- b) What is the turnaround time of each process of the scheduling algorithm in part a and also calculate Average turnaround time for each scheduling.
- c) What is waiting time of each process for each of process for each of the scheduling algorithm in part a & also calculate the average waiting time for each scheduling.
- d) Which of the scheduling in part a result in the minimal average waiting time (over all processes)?

OR

What advantages is there in having different time quantum sizes on different levels of a multilevel queuing system? Also find out the differences between multilevel queue scheduling and multilevel feedback scheduling.

Q.3(A) Explain spin-lock. Why spin-locks are not appropriate for uniprocessor systems yet may be suitable for multiprocessor systems. **03**

(B) Is it possible to have a deadlock involving only one process? Explain your answer. **03**

(C) Demonstrate the following algorithms. **06**

- a) The Bakery Algorithm.
- b) Peterson's Algorithm.

OR

Consider the following snapshot of a system. There are no current outstanding queued unsatisfied requests.

Available

r1 r2 r3 r4

2 1 0 0

Process	Current Allocation				Maximum Demand				still needs
r1 r2 r3 r4	r1	r2	r3	r4	r1	r2	r3	r4	
p1	0	0	1	2	0	0	1	2	
p2	2	0	0	0	2	7	5	0	
p3	0	0	3	4	6	6	5	6	
p4	2	3	5	4	4	3	5	6	
p5	0	3	3	2	0	6	5	2	

- a) What each process still might request and display in the columns labeled "still needs." Compute.
- b) Is the system currently in a safe or unsafe state? Explain.
- c) Is the System currently deadlocked? Why or why not? Explain.

Contd....

- d) Which Processes, if any, are or may become deadlocked? Explain.
- e) If a request from p3 arrives for (0,1,0,0), can that request be safely granted immediately ? In what state (deadlocked, safe, and unsafe) would immediately granting that whole request leave the system? Which process, if any, are or may become deadlocked if this whole request is granted is immediately?

- Q.4(A)** What is fragmentation? Differentiate between internal and external fragmentation. **03**
- (B)** What is the cause of thrashing? How does the system detect thrashing? **03**
- (C)** How many page faults occur for following algorithm for the following reference string, for four page frames? **06**
- 1,2,3,4,5,3,4,1,6,7,8,7,8,9,7,8,9,5,4,5,4,5,4,2
- a) LRU replacement
- b) FIFO replacement
- c) Optimal Replacement

OR

Differentiate Between :

- a) Paging and Segmentation.
- b) Virtual Memory and Cache memory.

- Q.5(A)** Give an example of an application in which data in a file should be accessed in the following order: **03**
- a) Sequentially
- b) Randomly.
- (B)** State three advantages of placing functionality in a device controller, rather than in the kernel. State three disadvantages. **03**
- (C)** Compare the performance of C-Scan and Scan scheduling with the help of Example. **06**

OR

Write short notes on following:

- a) Directory Structure
- b) File sharing.
- c) File System in Linux.

XXXXXX